⚡ Section: Electricity

🔷 Topic: Capacitors

(a)(i) Calculate the capacitance of the capacitor

✔3 marks

Given:

- $V = 5.7\,\mathrm{V}$
- $Q = 136.8\,\mathrm{mC} = 136.8 \times 10^{-3}\,\mathrm{C}$

Use the relationship:

$$C = \frac{Q}{V} = \frac{136.8 \times 10^{-3}}{5.7} = 0.0240\,\mathrm{F}$$

✅ Final answer:

$$24.0\,\mathrm{mF}$$

(a)(ii) Determine the absolute uncertainty in the capacitance

✔3 marks

Use percentage uncertainty method:

$$\text{Uncertainty in } Q = \frac{0.1}{136.8} \times 100 = 0.0731\% \quad \text{Uncertainty in } V = \frac{0.1}{5.7} \times 100 = 1.754$$

Total % uncertainty = $0.0731 + 1.754 = 1.827\%$

Now calculate absolute uncertainty:

$$\text{Absolute uncertainty} = \frac{1.827}{100} \times 0.0240 = 4.38 \times 10^{-4}\,\mathrm{F} = 0.00044\,\mathrm{F}$$

✅ Final answer:

$$(0.0240 \pm 0.0004)\,\mathrm{F}$$

(b) Calculate the estimated time taken for the capacitor to charge fully

✔2 marks

Use the relationship:

$t = 5RC$

Given:

- $R = 15\,\mathrm{k\Omega} = 15 \times 10^{3}\,\Omega$
- $C = 0.0240\,\mathrm{F}$

$$t = 5 \times 15 \times 10^{3} \times 0.0240 = 1800\,\mathrm{s}$$

🧠 Revision Tips

- Capacitors store charge: $Q = CV$
- Absolute uncertainty = (% uncertainty) × measured value
- Full charge time ≈ $5RC$, not just "RC"
- Work in SI units: F (farads), Ω (ohms), s (seconds)